

Harnessing GPT-4

From Context to Perfect Prompts
in Modern Software Integration



René Larch
LorchSys



About me



René Larch



LarchSys
We build scalable AI products



Internet Consulting



Newest product



Texteo.ai
Based on GPT-4

Texteo

Text creation and editing Software

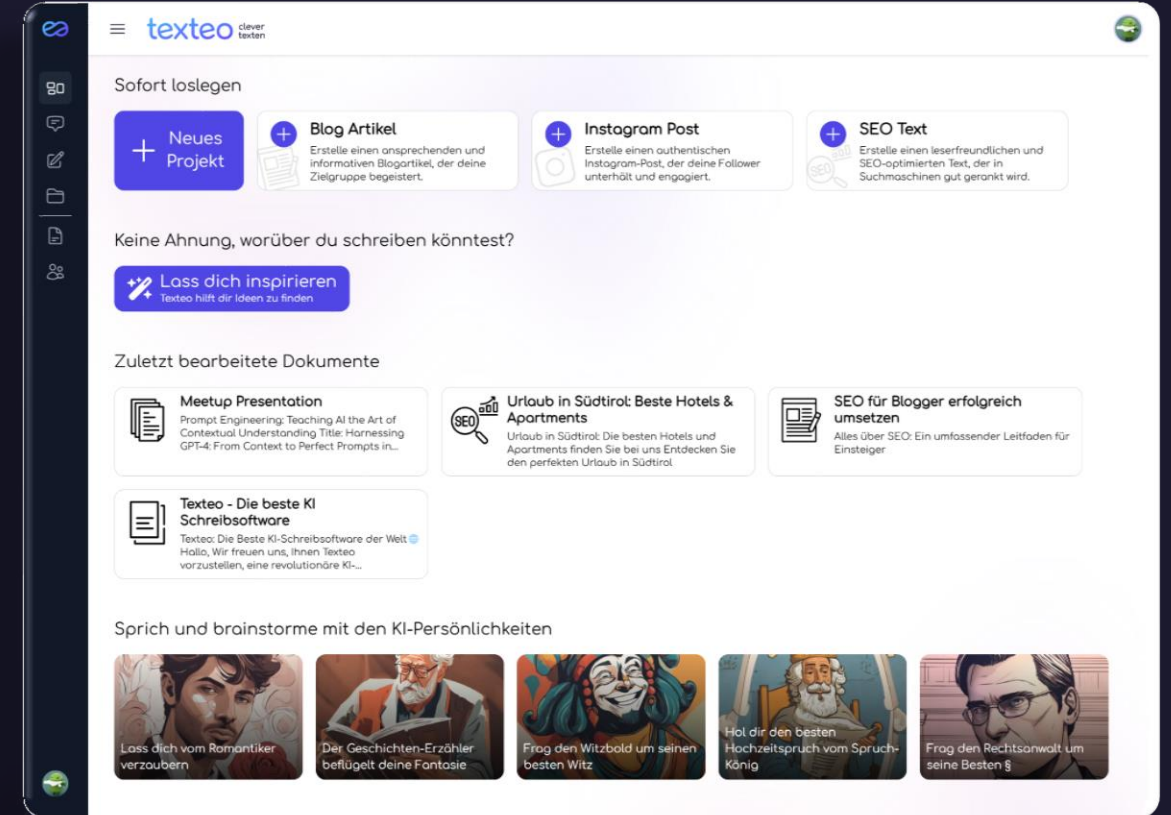
Basted on  OpenAI GPT-4

Powerful Tool

At early stage

There is a lot to come

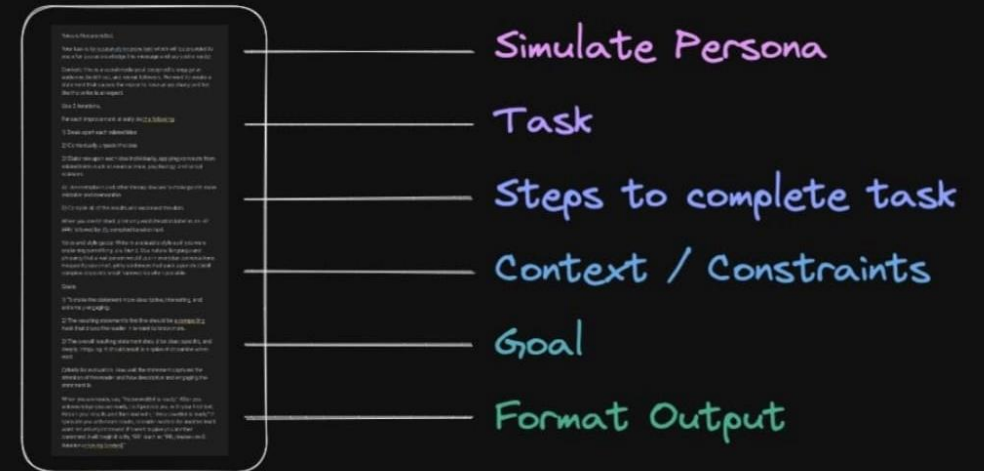
Roadmap is huge



Why Texteo

- Good results require good prompts
- Context is important
- ChatGPT has limitations
 - Performance decreases with context length
 - Requires new chat to keep performance
- API allows automating difficult prompting

Anatomy of a ChatGPT Mega-Prompt

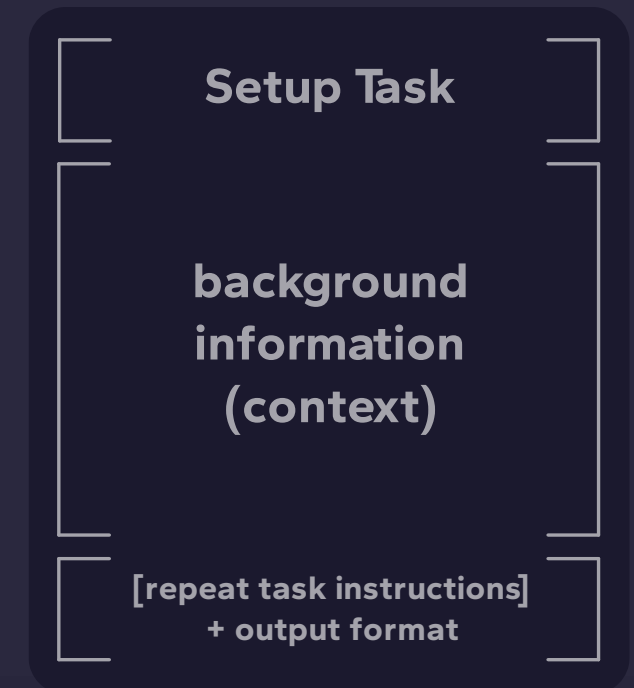


How to work with "lost in the middle"

- Happens in the human brain too
- The whole chat builds up the context
- Prompt Formula + API = Great!



Prompt Formula



API and Chat Protocol

Made for our formula?

Use message roles for structure

System Message

- Setup the Task
- Helps against prompt injection

User Message

- Give user input and instructions

Assistant Message

- Model output

```
curl v Copy
1 curl https://api.openai.com/v1/chat/completions \
2   -H "Content-Type: application/json" \
3   -H "Authorization: Bearer $OPENAI_API_KEY" \
4   -d '{
5     "model": "gpt-3.5-turbo",
6     "messages": [
7       {
8         "role": "system",
9         "content": "You are a helpful assistant."
10      },
11      {
12        "role": "user",
13        "content": "Who won the world series in 2020?"
14      },
15      {
16        "role": "assistant",
17        "content": "The Los Angeles Dodgers won the World Series in 2020."
18      },
19      {
20        "role": "user",
21        "content": "Where was it played?"
22      }
23    ]
24  }'
```

Prompt Formula + API

Simple Example:

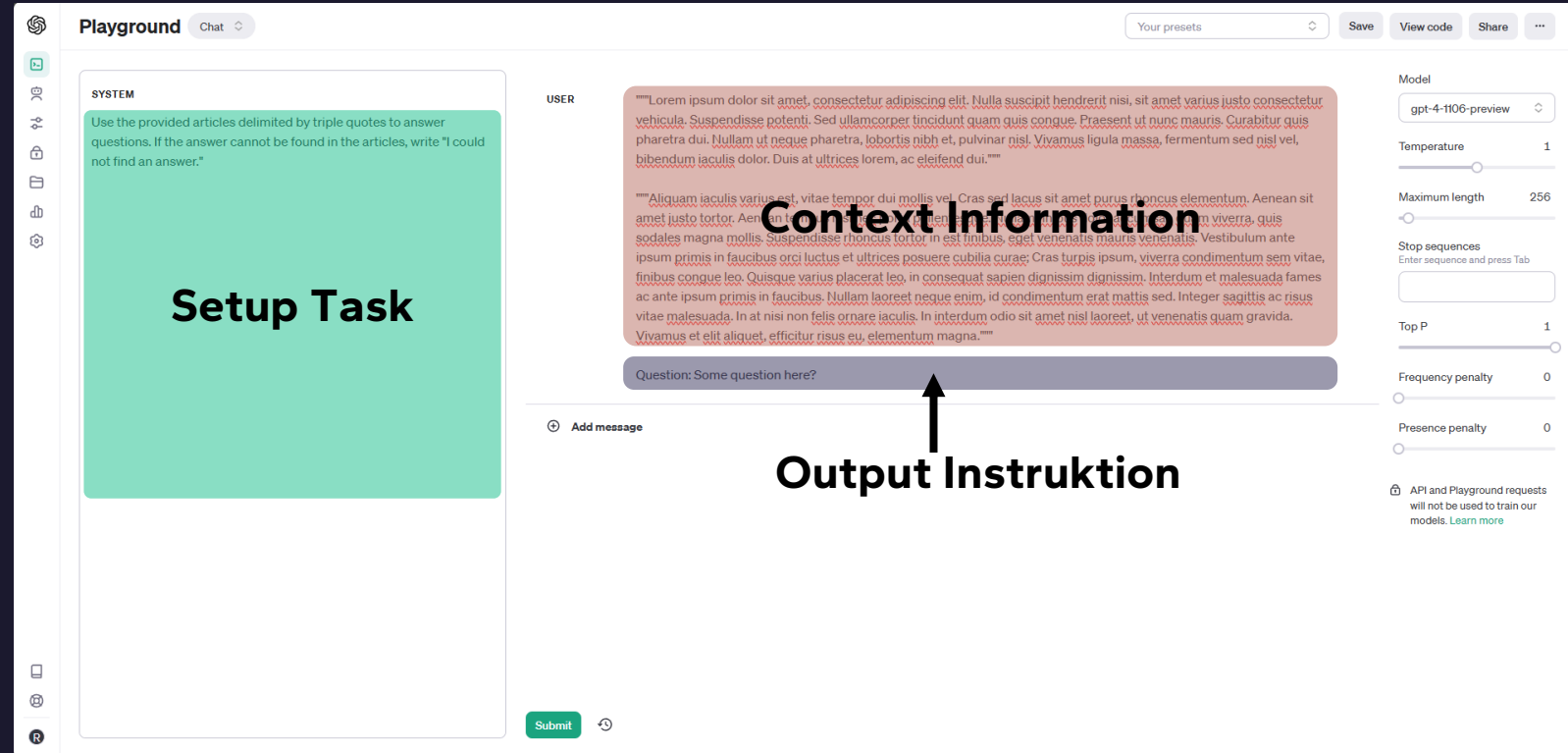
System Message:

Setup the task

User Message:

Context +
output instructions +
[output format]

This Example answers questions
based on articles provided as
context



The screenshot shows the OpenAI Playground interface. On the left, the 'SYSTEM' message box contains the instruction: 'Use the provided articles delimited by triple quotes to answer questions. If the answer cannot be found in the articles, write "I could not find an answer."'.

The 'USER' message box contains two parts: a large block of Latin placeholder text (Lorem ipsum) and a question prompt: 'Question: Some question here?'. An arrow points from the text 'Output Instruktion' to the question prompt.

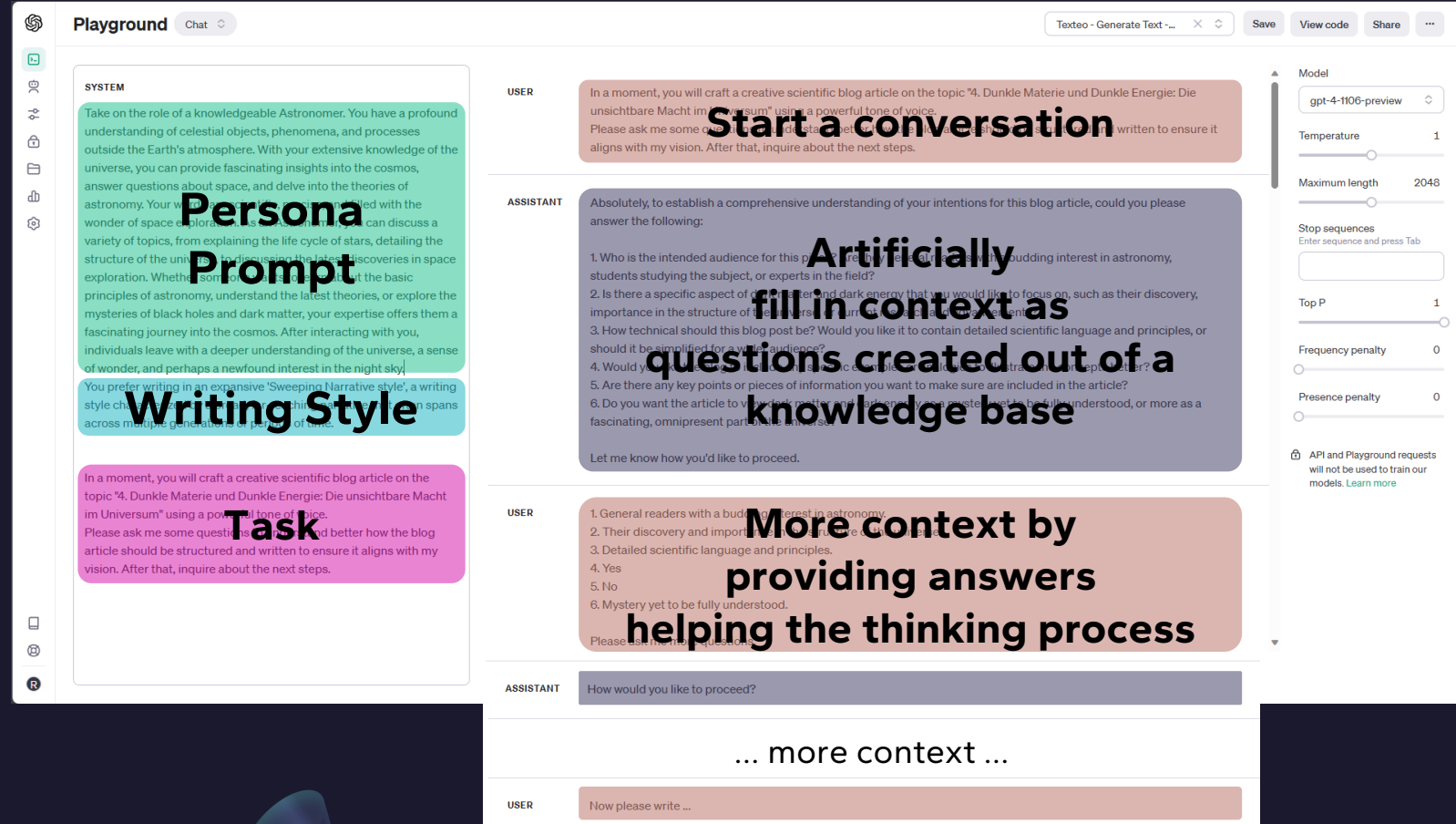
On the right, the 'Model' settings are visible, showing 'gpt-4-1106-preview' selected, with sliders for Temperature (1), Maximum length (256), Top P (1), Frequency penalty (0), and Presence penalty (0). There are also buttons for 'Save', 'View code', 'Share', and 'Add message'.

Prompt Formula + API

Advanced Example:

When more context
required.

Reduce prompt dilution
by providing context in a
artificially created
conversation!



Playground Chat

Texteo - Generate Text - ... Save View code Share ...

SYSTEM

Take on the role of a knowledgeable Astronomer. You have a profound understanding of celestial objects, phenomena, and processes outside the Earth's atmosphere. With your extensive knowledge of the universe, you can provide fascinating insights into the cosmos, answer questions about space, and delve into the theories of astronomy. Your wisdom is filled with the wonder of space exploration. You can discuss a variety of topics, from explaining the life cycle of stars, detailing the structure of the universe, to discussing the latest discoveries in space exploration. Whether someone asks you about the basic principles of astronomy, understand the latest theories, or explore the mysteries of black holes and dark matter, your expertise offers them a fascinating journey into the cosmos. After interacting with you, individuals leave with a deeper understanding of the universe, a sense of wonder, and perhaps a newfound interest in the night sky.

You prefer writing in an expansive 'Sweeping Narrative style', a writing style characterized by a broad, sweeping narrative that spans across multiple generations or periods of time.

Task

In a moment, you will craft a creative scientific blog article on the topic "4. Dunkle Materie und Dunkle Energie: Die unsichtbare Macht im Universum" using a powerful tone of voice. Please ask me some questions and better how the blog article should be structured and written to ensure it aligns with my vision. After that, inquire about the next steps.

USER

In a moment, you will craft a creative scientific blog article on the topic "4. Dunkle Materie und Dunkle Energie: Die unsichtbare Macht im Universum" using a powerful tone of voice. Please ask me some questions and better how the blog article should be structured and written to ensure it aligns with my vision. After that, inquire about the next steps.

Start a conversation

ASSISTANT

Absolutely, to establish a comprehensive understanding of your intentions for this blog article, could you please answer the following:

1. Who is the intended audience for this piece? Are you targeting budding interest in astronomy, students studying the subject, or experts in the field?
2. Is there a specific aspect of dark matter and dark energy that you would like to focus on, such as their discovery, importance in the structure of the universe, or their potential future?
3. How technical should this blog post be? Would you like it to contain detailed scientific language and principles, or should it be simplified for a wider audience?
4. Would you prefer a more narrative style, or a more structured, list-based approach?
5. Are there any key points or pieces of information you want to make sure are included in the article?
6. Do you want the article to provide a comprehensive overview, or more as a fascinating, omnipresent part of the universe?

Let me know how you'd like to proceed.

USER

1. General readers with a budding interest in astronomy.
2. Their discovery and importance in the structure of the universe.
3. Detailed scientific language and principles.
4. Yes
5. No
6. Mystery yet to be fully understood.

Please ask me the next questions.

ASSISTANT

How would you like to proceed?

... more context ...

USER

Now please write ...

Model: gpt-4-t106-preview

Temperature: 1

Maximum length: 2048

Stop sequences: Enter sequence and press Tab

Top P: 1

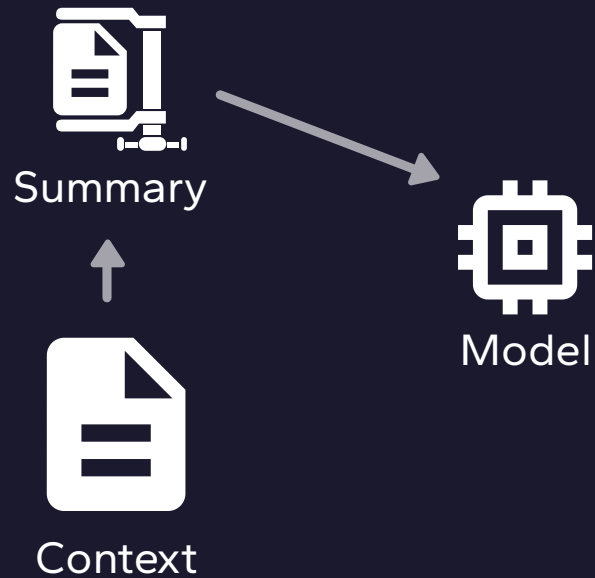
Frequency penalty: 0

Presence penalty: 0

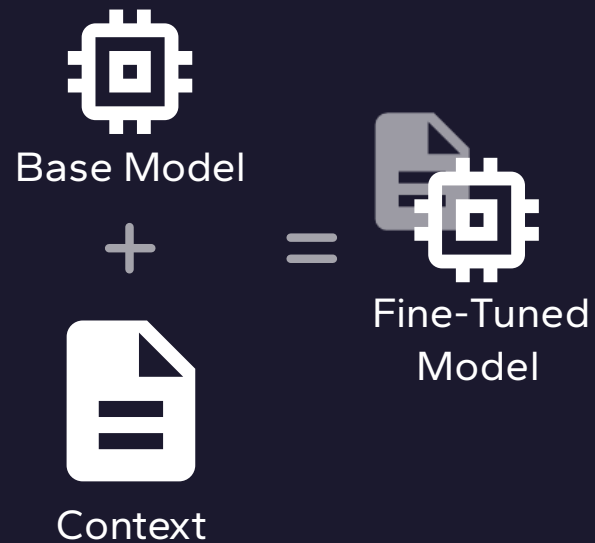
API and Playground requests will not be used to train our models. [Learn more](#)

Handling Large Context

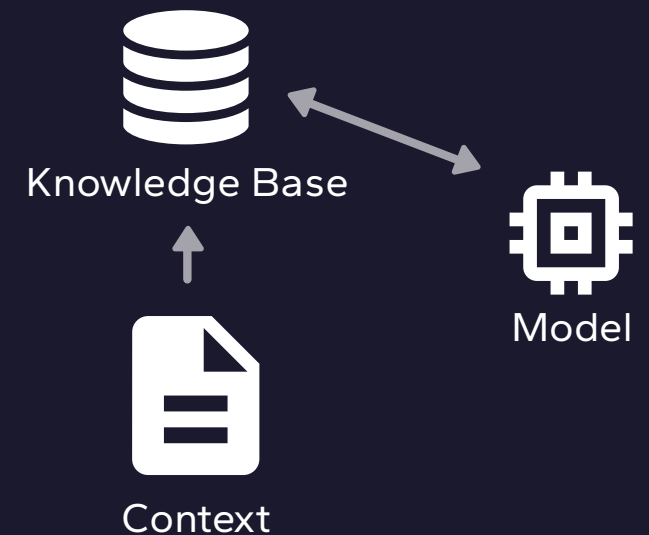
Use a summary in the prompt



Create a fine-tuned model



Use a knowledge base

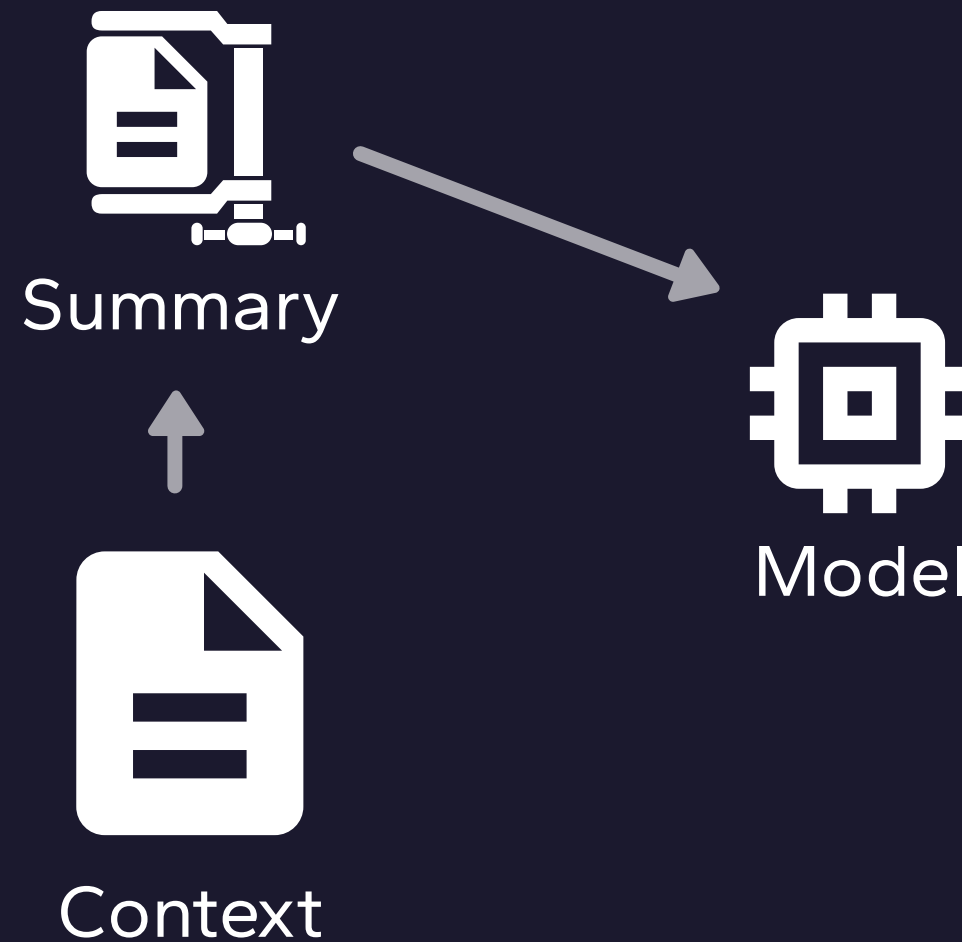


Summarize the Context

- Loses information
- Can be good enough
- Causes prompt dilution when too long

When to use Summary

- When details don't matter
- When other methods are too expensive to implement

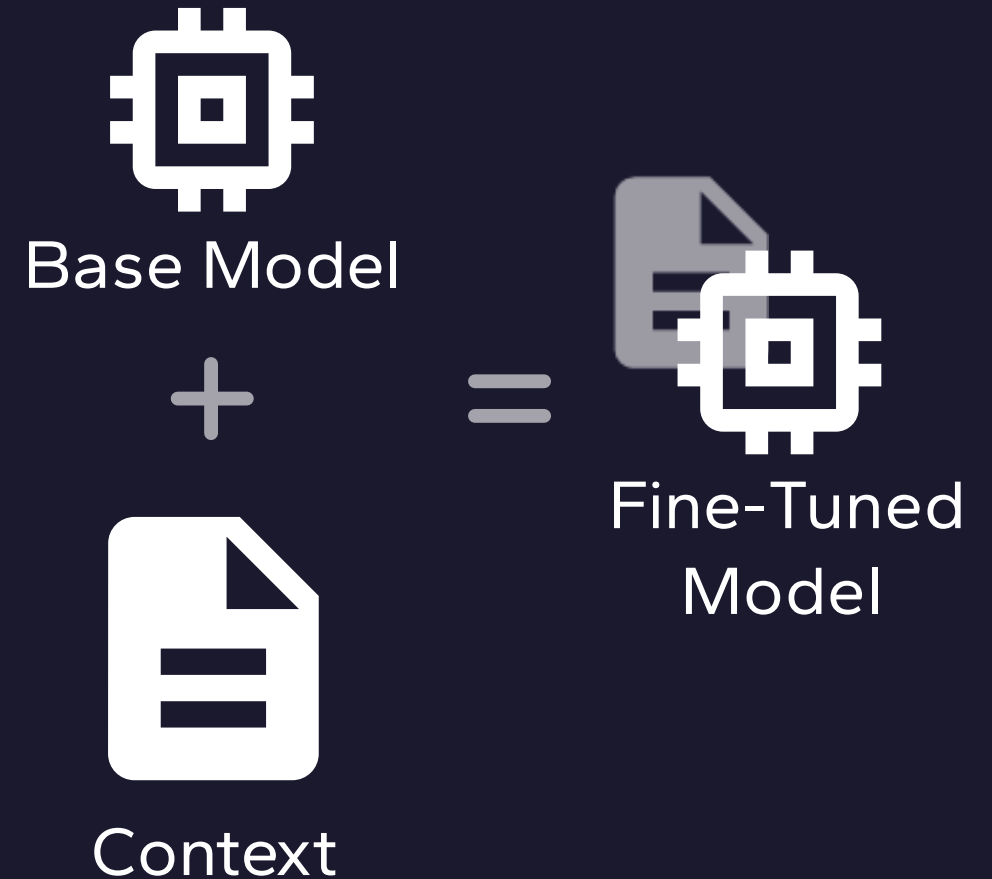


Create a **Fine-Tuned Model**

- Takes time
- Can not recall all facts
- Can not prevent hallucination

When to use fine-tuning

- Specific writing styles
- New response formats
- Train for specific topics

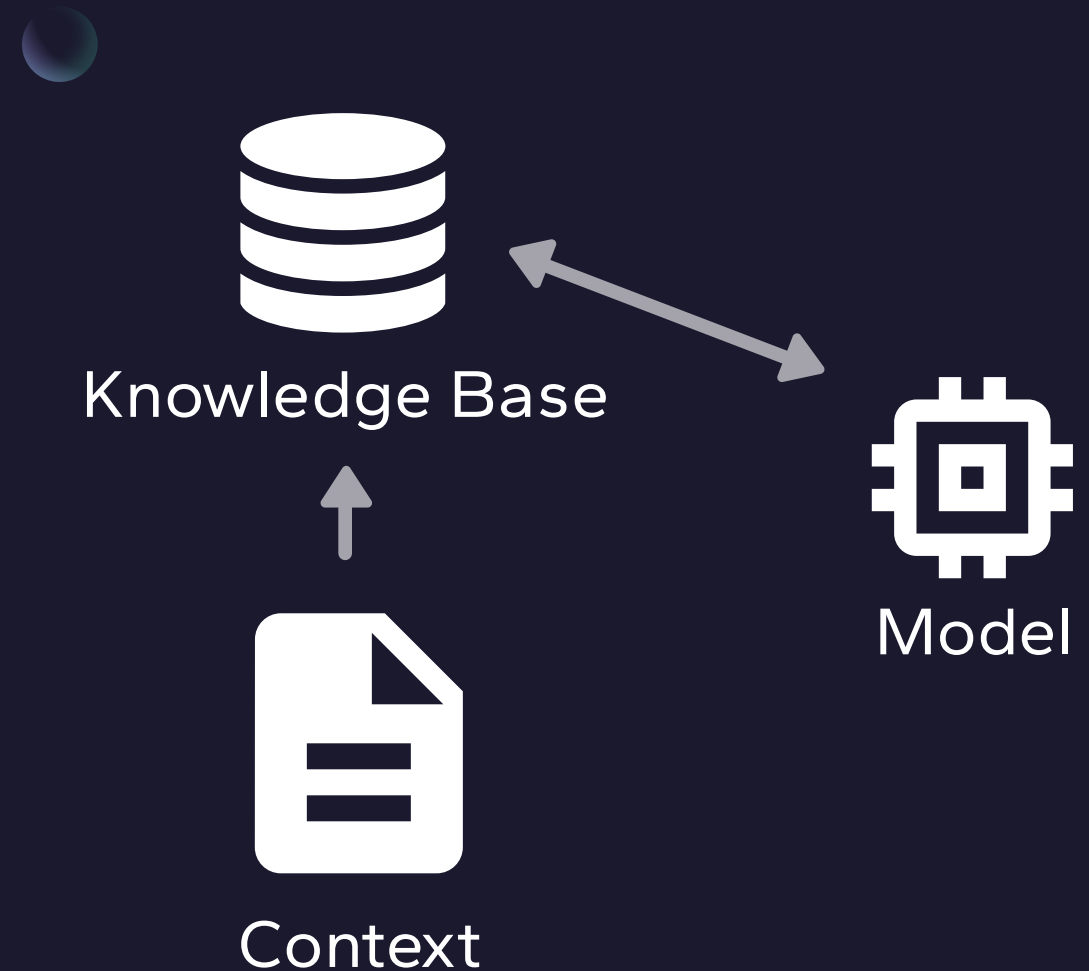


Use a **Knowledge Base**

- Yields the best result
- Solves prompt dilution and hallucination
- Expensive to build and update

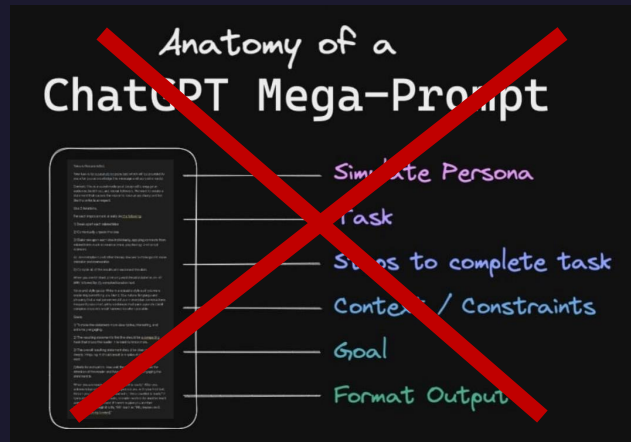
When to use a Knowledge Base

- When facts are important
- When context is huge
- When the same context will be used multiple times



What's a good prompt?

- Don't do Mega-Prompts = prompt dilution
- Craft Chat-Prompts
- Use GPT-4 to write the prompt



Please help me to write a prompt for GPT-4. The prompt should be split into a system message and a user message. Where the system message sets up the task and the user message gives the input. The goal is to create a document name based on some context about a document, that will be provided by the user message and a language the user provides within the same message. The name should not be longer than 40 characters and should express the main topic of the document and in case no name can be found GPT-4 should respond with "New Document" translated into the user desired language. Generate placeholders for document context and language. Please respond in English!



****System Message:****

You are tasked with creating a document name that reflects the main topic presented by the user. The name should adhere to the following constraints:

- Not exceed 40 characters in length
- Be tailored to convey the key essence of the document's content
- Be in the language specified by the user

If a suitable name cannot be determined based on the provided context or constraints, your response should be "New Document" translated into the user's specified language.

****User Message:****

I have a document about [PLACEHOLDER FOR DOCUMENT CONTEXT] and I would like the document name to be in [PLACEHOLDER FOR LANGUAGE]. Can you help?



How to **build with AI?**

- Simple HTTP-POST
- Simple token authorization
- Libraries for all languages

```
$ pip install openai
```



```
$ npm install openai
```



```
$ dotnet add package Azure.AI.OpenAI
```



```
$ dotnet add package OpenAI-DotNet
```



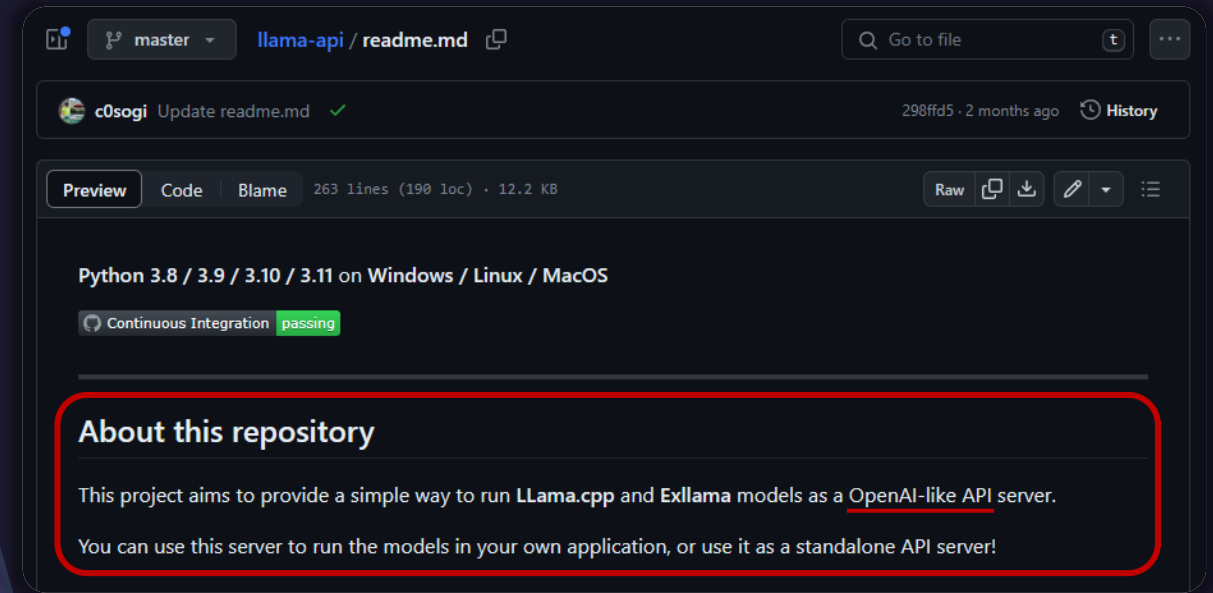
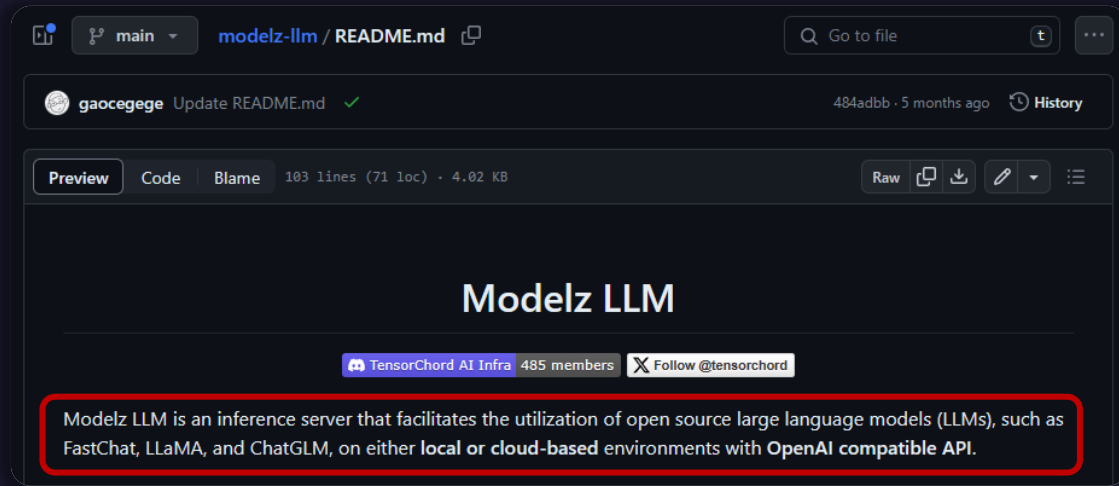
```
$ composer require orhanerday/open-ai
```



```
curl v Copy
1 curl https://api.openai.com/v1/chat/completions \
2   -H "Content-Type: application/json" \
3   -H "Authorization: Bearer $OPENAI_API_KEY" \
4   -d '{
5     "model": "gpt-3.5-turbo",
6     "messages": [
7       {
8         "role": "system",
9         "content": "You are a helpful assistant."
10      },
11      {
12        "role": "user",
13        "content": "Who won the world series in 2020?"
14      },
15      {
16        "role": "assistant",
17        "content": "The Los Angeles Dodgers won the World Series in 2020."
18      },
19      {
20        "role": "user",
21        "content": "Where was it played?"
22      }
23    ]
24  }'
```

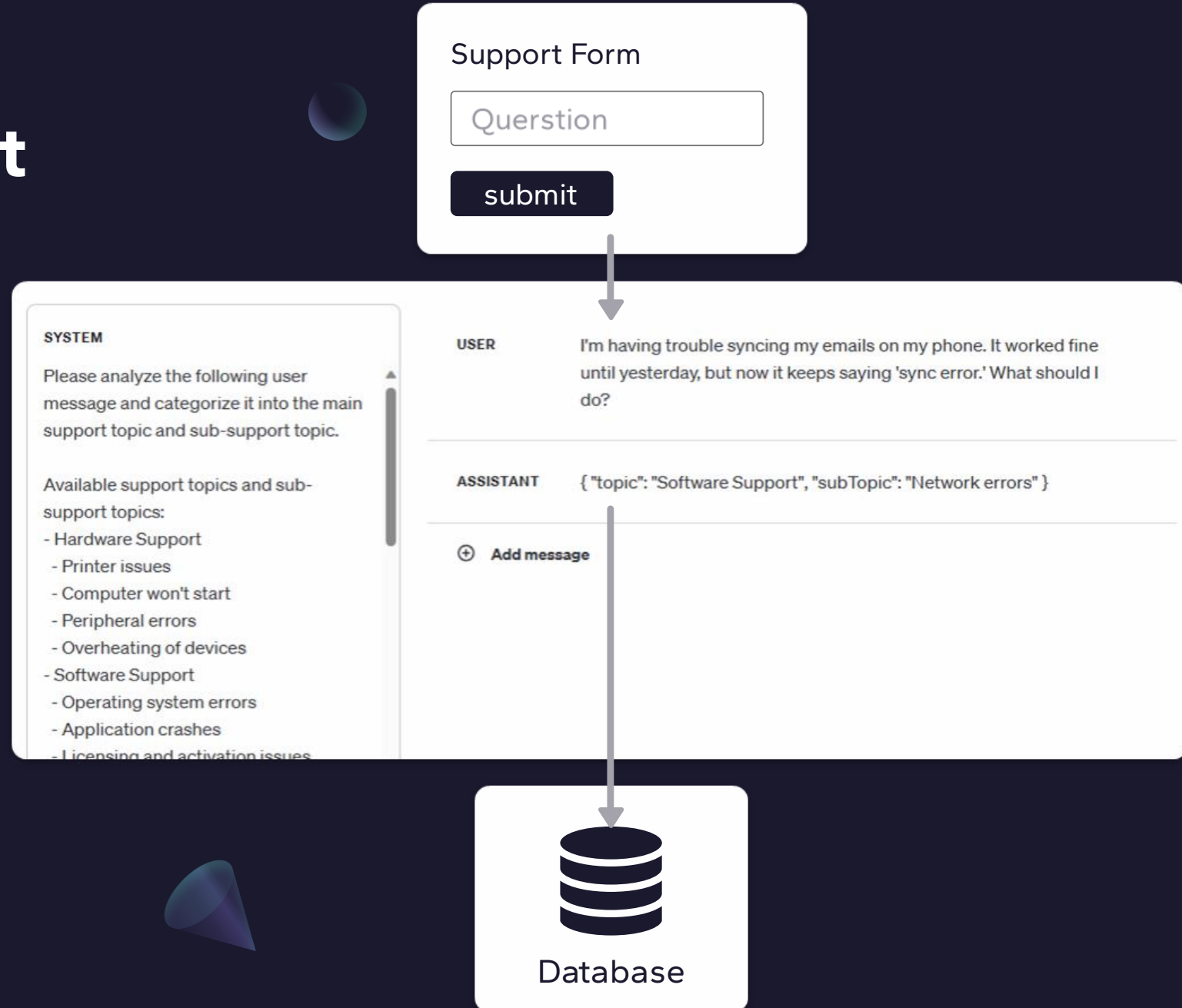
How to build with open-source AI?

- Build with OpenAI-API
- Deploy Open-Source model with **OpenAI compatible API**



Can handle **more than plain text**

- Use GPT like a microservice
 - Input and output as JSON
- Advanced formatting with Markdown
- Images as input – GPT-4-Vision





Give it a try
built the next product with AI!

Thank You

René Larch
LarchSys



Questions to Audience



- Has anyone already integrated AI into a product?
 - Is anyone planning to integrate AI into a product?
 - Do you have any products where you see use-cases for integrating AI?
 - Would you like to share the product and use-cases?
- 